

Regression Modeling Strategies: A Comparative Investigation of Linear, Polynomial, and Regularized Approaches

Anshika Yadav (24B0326)

Mentor: Prince

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Executive Overview

This document presents findings from an empirical investigation into four regression methodologies applied to real estate valuation. The study employs the Boston Housing dataset to systematically evaluate how model architecture decisions—ranging from simple linear assumptions to polynomial expansions and penalty-based regularization—influence predictive accuracy and generalizability.

Through rigorous cross-validation and hyperparameter optimization, we demonstrate that while second-degree polynomial expansions yield superior training performance, regularized models, particularly Ridge regression, provide more reliable generalization to unseen observations. The analysis further reveals that Lasso regularization effectively identifies the most influential predictors, with average room count and neighborhood socioeconomic status emerging as dominant factors in property valuation.

1 Research Context and Objectives

1.1 Foundational Challenge

The central problem in supervised learning involves developing models that achieve strong performance on both training data and previously unseen examples. This tension, formalized through the bias-variance decomposition, represents one of the most persistent challenges in statistical learning theory. Regression problems, where the target variable is continuous, exemplify this challenge—simple parametric forms may fail to capture underlying patterns, while overly flexible models risk memorizing noise rather than learning signal.

1.2 Practical Significance

Regression modeling finds application across numerous domains where numerical prediction is essential. In real estate markets, accurate valuation models inform investment decisions, mortgage underwriting, and urban planning. The Boston Housing dataset, despite its age, provides a well-characterized testbed for evaluating regression techniques due to its mix of continuous and categorical predictors and known relationships among variables.

1.3 Investigation Aims

The present study addresses five interconnected questions:

1. What is the relationship between model complexity—operationalized through polynomial degree—and performance across training and test partitions?
2. How do L1 and L2 regularization strategies constrain model complexity and enhance generalization?
3. Which modeling approach achieves the optimal tradeoff between systematic error and sensitivity to training data fluctuations in the Boston Housing context?

4. What is the impact of feature standardization on regularized model performance?
5. Which features contribute most substantially to housing price predictions, and how does Lasso facilitate feature selection?

1.4 Analytical Boundaries

The investigation encompasses:

- Comprehensive exploratory analysis with visualization of variable distributions and interrelationships
- Systematic preprocessing including standardization and partitioning
- Training and evaluation of four model classes
- Hyperparameter optimization through cross-validation
- Comparative performance assessment using multiple metrics
- Feature importance interpretation through coefficient analysis

2 Theoretical Foundations

2.1 Supervised Learning Paradigm

Supervised learning constitutes a machine learning framework wherein algorithms infer mappings from labeled training examples to generate predictions for new inputs. Within this paradigm, regression tasks involve continuous target variables, necessitating optimization of loss functions that penalize prediction errors. The learning process fundamentally involves minimizing some measure of discrepancy between predicted and actual values.

2.2 Bias-Variance Decomposition

The bias-variance tradeoff provides a theoretical framework for understanding model generalization. Bias quantifies error introduced by approximating complex real-world relationships with simplified functional forms—high bias manifests as systematic underfitting. Variance captures model sensitivity to training data fluctuations—high variance leads to overfitting, where random noise is incorporated into the learned function.

The expected prediction error decomposes as:

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \text{Bias}[\hat{f}(\mathbf{x})]^2 + \text{Var}[\hat{f}(\mathbf{x})] + \sigma^2 \quad (1)$$

This decomposition guides model selection: the goal is to identify the configuration that minimizes total expected error by balancing systematic and random components.

2.3 Model Evaluation Strategies

Reliable performance estimation requires separating data into training, validation, and test partitions. Cross-validation enhances robustness by averaging performance across multiple data splits. k -fold cross-validation involves partitioning data into k subsets, iteratively training on $k - 1$ subsets and validating on the held-out portion.

2.4 Candidate Model Classes

2.4.1 Linear Regression

The linear model posits a straight-line relationship between predictors and response:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_d x_d + \epsilon \quad (2)$$

Parameter estimation via ordinary least squares minimizes the sum of squared residuals:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 \quad (3)$$

While interpretable and computationally efficient, linear models may underperform when relationships are inherently nonlinear.

2.4.2 Polynomial Extensions

Polynomial regression augments the feature space with higher-order terms:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d + \epsilon \quad (4)$$

This approach captures curvature while maintaining linearity in parameters. However, increasing polynomial degree expands model complexity, potentially inducing overfitting.

2.4.3 Regularized Formulations

Regularization introduces penalty terms to constrain coefficient magnitudes:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \cdot \text{Penalty}(\beta) \quad (5)$$

Ridge (L2 Penalty): Adds squared coefficient sum:

$$\hat{\beta}^{\text{Ridge}} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \sum_{j=1}^d \beta_j^2 \quad (6)$$

Coefficients are shrunk toward zero but rarely reach exactly zero, preserving all features.

Lasso (L1 Penalty): Adds absolute coefficient sum:

$$\hat{\beta}^{\text{Lasso}} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \sum_{j=1}^d |\beta_j| \quad (7)$$

Lasso can drive coefficients to exactly zero, enabling automatic feature selection.

2.5 Performance Metrics

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (8)$$

Root Mean Squared Error (RMSE) provides interpretability in original units.

R-squared:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (9)$$

While R^2 values approaching 1 indicate better fit, this metric can be misleading as it tends to increase with model complexity.

3 Experimental Design and Implementation

3.1 Dataset Characteristics

The Boston Housing dataset, originally compiled by Harrison and Rubinfeld (1978), contains 506 observations with 13 predictor variables describing housing characteristics across Boston suburbs. The target variable, MEDV, represents median home value in thousands of dollars.

Table 1: Feature Descriptions

Variable	Description
CRIM	Per capita crime rate by town
ZN	Proportion of residential land zoned for large lots
INDUS	Proportion of non-retail business acres
CHAS	Charles River dummy (1 if adjacent)
NOX	Nitric oxides concentration (parts per 10 million)
RM	Average rooms per dwelling
AGE	Proportion of owner-occupied units built before 1940
DIS	Weighted distance to employment centers
RAD	Radial highway accessibility index
TAX	Full-value property tax rate per \$10,000
PTRATIO	Pupil-teacher ratio by town
B	$1000(B_k - 0.63)^2$ where B_k is proportion of Black residents
LSTAT	Percentage of lower-status population

3.2 Preprocessing Protocol

Data Quality Assessment: Initial inspection confirmed no missing values. Outlier detection employed z-score analysis (threshold = 3) and interquartile range methods.

Feature Standardization: StandardScaler was applied to transform features to zero mean and unit variance:

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma} \quad (10)$$

This step is critical for regularized models where coefficient magnitudes are penalized uniformly.

Data Partitioning: The dataset was split into 80% training (404 samples) and 20% testing (102 samples) using stratified random sampling with a fixed seed for reproducibility.

3.3 Computational Environment

The analysis was conducted using Python 3.8 with the following libraries:

- NumPy (1.21.0): Numerical operations
- Pandas (1.3.0): Data manipulation
- Matplotlib (3.4.0): Visualization
- Seaborn (0.11.0): Statistical visualization
- Scikit-learn (0.24.0): Machine learning implementations

Table 2: Scikit-learn Functions Employed

Function	Purpose
LinearRegression	Ordinary least squares
PolynomialFeatures	Generate polynomial terms
Ridge	L2-regularized regression
Lasso	L1-regularized regression
cross_val_score	Cross-validation performance
GridSearchCV	Hyperparameter optimization

3.4 Experimental Workflow

The investigation proceeded through:

1. Data preprocessing and standardization
2. Model training on training partition
3. Hyperparameter tuning via 5-fold cross-validation

4. Test set evaluation using MSE, RMSE, and R^2
5. Comparative performance analysis
6. Feature importance interpretation

4 Empirical Findings

4.1 Comparative Performance Assessment

Table 3: Model Performance Metrics

Model Configuration	RMSE	Test R^2	Training R^2	Overfit Gap
Linear (OLS)	4.824	0.732	0.741	0.009
Polynomial (deg 2)	4.213	0.796	0.852	0.056
Polynomial (deg 4)	5.891	0.601	0.918	0.317
Ridge ($\alpha = 1.0$)	4.789	0.735	0.740	0.005
Ridge ($\alpha = 10.0$)	4.802	0.733	0.738	0.005
Lasso ($\alpha = 0.01$)	4.815	0.733	0.739	0.006
Lasso ($\alpha = 0.1$)	4.893	0.724	0.732	0.008

4.2 Linear Model Performance

Ordinary least squares regression achieved test $R^2 = 0.732$, indicating that approximately 73% of variance in housing values is captured by linear relationships. The RMSE of 4.824 (\$4,824 in thousands) serves as a baseline for comparison. The narrow gap between training and test performance (0.009) suggests minimal overfitting.

Table 4: Linear Model Coefficients

Feature	Coefficient	Feature	Coefficient
RM	3.246	CRIM	-0.984
LSTAT	-2.754	CHAS	0.654
PTRATIO	-2.149	NOX	-0.521
DIS	1.876	AGE	0.312

4.3 Polynomial Model Analysis

Introducing second-degree polynomial terms improved performance substantially, with test R^2 reaching 0.796 and RMSE decreasing to 4.213. This suggests that nonlinear relationships between predictors and housing values are present in the data.

However, fourth-degree polynomial expansion led to severe overfitting. While training R^2 increased to 0.918, test performance degraded to $R^2 = 0.601$ with RMSE rising to 5.891. This pattern exemplifies the bias-variance tradeoff—excessive complexity enables the model to capture noise rather than generalizable patterns.

4.4 Regularized Model Evaluation

Ridge regression with $\alpha = 1.0$ achieved test $R^2 = 0.735$ with RMSE = 4.789, slightly outperforming ordinary least squares. The training-test gap was minimal (0.005), indicating effective complexity control.

Lasso regression with $\alpha = 0.01$ yielded comparable performance (test $R^2 = 0.733$, RMSE = 4.815). Increasing regularization strength to $\alpha = 0.1$ led to modest performance degradation, suggesting that some shrinkage may bias the model excessively.

4.5 Feature Selection Outcomes

Table 5: Lasso Coefficient Sparsity ($\alpha = 0.01$)

Retained Features	Zeroed Features
RM (2.987)	AGE (0.000)
LSTAT (-2.512)	INDUS (0.000)
PTRATIO (-1.876)	ZN (0.000)
DIS (1.543)	RAD (0.000)
CRIM (-0.876)	B (0.000)
CHAS (0.543)	
NOX (-0.412)	

Lasso effectively eliminated five features, identifying RM, LSTAT, PTRATIO, DIS, CRIM, CHAS, and NOX as the most influential predictors. This sparsity pattern provides insight into which variables drive housing prices in the Boston market.

5 Interpretation and Discussion

5.1 Bias-Variance Dynamics

The results illustrate the bias-variance tradeoff in concrete terms. Linear regression exhibits relatively high bias (assuming linearity) but low variance (stable across data partitions). Second-degree polynomial reduces bias by capturing curvature while maintaining reasonable variance. Fourth-degree polynomial dramatically reduces bias on training data but introduces excessive variance, leading to poor generalization.

Regularization strategies effectively navigate this tradeoff. Ridge regression, by uniformly shrinking coefficients, reduces variance while incurring modest bias. Lasso achieves similar effects while additionally performing feature selection, which can enhance interpretability.

5.2 Feature Importance Analysis

The coefficient patterns reveal meaningful economic relationships:

- **Average room count (RM):** Strong positive association—larger homes command premium prices

- **Lower-status population proportion (LSTAT):** Strong negative association—neighborhood socioeconomic composition significantly impacts valuation
- **Pupil-teacher ratio (PTRATIO):** Negative relationship—better schools (lower ratios) increase property values
- **Distance to employment centers (DIS):** Positive relationship—accessibility enhances desirability
- **Crime rate (CRIM):** Negative relationship—safety concerns depress prices
- **Charles River proximity (CHAS):** Positive relationship—waterfront locations command premiums
- **Nitric oxide concentration (NOX):** Negative relationship—air quality affects valuations

Features eliminated by Lasso (AGE, INDUS, ZN, RAD, B) appear to have limited marginal predictive value once other variables are considered, suggesting they may proxy for information captured by other features or have genuinely weak relationships with housing prices.

5.3 Regularization Impact Assessment

Both Ridge and Lasso effectively controlled overfitting, as evidenced by minimal training-test performance gaps. The choice between them depends on priorities:

- **Ridge:** Preferable when all features are believed to have some predictive value and multicollinearity is present
- **Lasso:** Advantageous when feature selection is desired for interpretability or when many features may be irrelevant

The optimal regularization strength (α) requires careful tuning, as excessive penalization can induce harmful bias. Cross-validation provides a principled approach to this selection.

5.4 Study Limitations

Several constraints warrant acknowledgment:

1. **Temporal scope:** Data from 1978 may not reflect contemporary housing markets
2. **Geographic specificity:** Findings may not generalize to other regions or countries
3. **Model scope:** Additional techniques (elastic net, Bayesian regression, gradient boosting) were not examined
4. **Feature engineering:** Transformations beyond polynomials (log, interaction terms) were not explored
5. **Ethical considerations:** The B variable's inclusion raises concerns about reinforcing historical biases

6 Conclusions and Future Directions

6.1 Key Findings Summary

This investigation yields several conclusions:

1. Simple linear regression provides a reasonable baseline ($R^2 = 0.732$) with excellent interpretability and minimal overfitting
2. Second-degree polynomial expansion captures meaningful nonlinearity, achieving the best test performance ($R^2 = 0.796$)
3. Excessive polynomial complexity induces overfitting, with test performance deteriorating substantially
4. Regularization techniques effectively balance bias and variance, with Ridge achieving competitive results and Lasso enabling feature selection
5. Room count and socioeconomic status emerge as the primary drivers of housing valuation

6.2 Practical Implications

For practitioners, these findings suggest:

- Model selection should be guided by validation performance rather than training performance
- Regularization provides value even when overfitting is not immediately evident
- Feature importance analysis can reveal substantive relationships in the data
- Cross-validation is essential for reliable model comparison and hyperparameter selection

6.3 Proposed Extensions

Building on this foundation, the following research directions are planned:

1. **Unsupervised learning:** Exploration of clustering and PCA techniques
2. **Alternative learners:** Implementation of support vector machines and decision trees
3. **Ensemble methods:** Investigation of random forests and gradient boosting
4. **Additional datasets:** Application to contemporary housing data from multiple regions
5. **End-to-end pipeline:** Development of a complete modeling workflow from problem formulation to final deployment

The analytical framework established in this study will inform subsequent investigations and contribute to a comprehensive understanding of regression modeling strategies.

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